Assignment 5 — Beta Build: Rendering integrated into the core loop

Minh Quan Dao (qd66) · Interactive Diffusion-Based 3D Generation Studio · Pillars: **Procedural Generation** (core) + **Rendering** (secondary) · Beta **v0.3.0**

Play: live studio (lab A100 server; quick-tunnel URL) · static build (viewer, gallery, exhibition; no GPU) · **Build zip:** v0.3.0-beta.zip · **Repo:** github.com/quandao10/diffusion-3d-studio · A3 video

1. Quick review: pillar 1 (procedural generation) is stable

Text → 3D Gaussians optimised by score distillation from a frozen diffusion prior (DreamGaussian; MVDream multi-view guidance by default, 500 steps). The A4 game loop is unchanged in structure: **menu → Challenge (describe a secret object without its taboo words) → GPU job with queue, ETA, Stop → result**. For this Beta, 6 real Challenge rounds were played through the public URL at the same time (one per session, MVDream, 6 free A100s): all 6 jobs finished, 166–166 s per round from submit to judged model (queue, model load, 500 steps, views, judge); the 500 optimisation steps took 89–92 s, the same as a single job in A3 (89–94 s), so parallel players do not slow each other down. These rounds fix the A4 known issue “no real GPU Challenge round recorded yet”. Raw data: `evidence/beta_rounds.json`.

2. Secondary pillar integration: the player's rendering decides the round

1 · Describe. Secret target + taboo words; prompt checked in the browser and on the server.

2 · Generate. 3D Gaussians on a free GPU; live loss and ETA. Then a *preview* verdict from 8 automatic views.

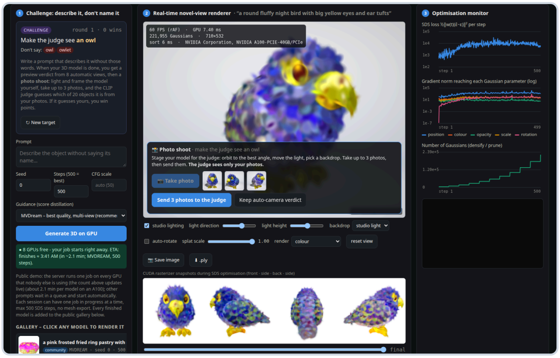
3 · Photo shoot (new). Light, frame and photograph the model in the real-time renderer; up to 3 photos.

4 · Verdict. The CLIP judge sees **only your photos**: win = target ranked 1st of 20; points = 100 × its probability. Wins go to the exhibition.

- **Lit Gaussian splatting.** Each Gaussian's normal is its shortest axis (the flat direction of the splat), flipped to point away from the object's centre, octahedral-encoded into the spare channel of my packed texture (no extra memory). The vertex shader adds wrap diffuse, Blinn specular and a cool rim light from a movable key light, so the cost is per Gaussian, not per pixel.
- **Stage and soft shadow.** Gradient backdrop and a ground disc under the model. The shadow is computed by projecting every Gaussian centre along the light onto the floor, accumulating opacity in a 160^2 grid (256^2 in the exhibition), blurring it and uploading it as a texture; it is only recomputed when the light or the model changes.
- **Why it is part of the loop, not a demo:** the same model scores differently depending on how it is rendered. Same 3 camera angles per round, 4 looks, judged by the same CLIP judge:

target	player prompt (no taboo words)	auto preview (8 views)	raw, white	raw, black	studio dark	studio light
owl	*“a round fluffy night bird with big yellow eyes and ear tufts”*	76%	80%	90%	97%	93%
pineapple	*“a spiky tropical fruit with a crown of green leaves”*	42% ✗	36% ✗	48% ✗	46% ✗	22% ✗
hamburger	*“a grilled beef patty sandwich with lettuce, tomato and cheese in a sesame bun”*	100%	100%	100%	100%	100%
lighthouse	*“a tall red and white striped coastal tower with a lamp room on top”*	57%	28% ✗	55%	69%	79%
penguin	*“a black and white flightless antarctic bird standing upright”*	66%	91%	66%	70%	66%
donut	*“a pink frosted fried ring pastry with rainbow sprinkles”*	99%	100%	99%	94%	86%
wins / mean target probability		5/6 · 0.73	4/6 · 0.72	5/6 · 0.76	5/6 · 0.80	5/6 · 0.74

Target probability (✗ = judge picked another object). Photos taken by a scripted player through the real UI (front, ±35°), `scripts/shoot_rounds.py` → `evidence/beta_shoot.csv`. Studio lighting on the dark stage raised the mean target probability from 0.72 (raw splats on white) to 0.80 and turned a lighthouse from a loss into a win. It is not always better (a penguin, a donut scored highest raw on white), so the look is a real choice for the player. The rounds themselves were played in the studio-light look: 425 pts · 5/6 wins · streak 4.


1 Challenge: describe it, don't name it
CHALLENGE
round 1 · 0 wins
Make the judge see an owl
Don't say: owl owlet
New target
Prompt
Describe the object without saying its name...
Seed
0
Steps (500 = best)
500
CFG scale
auto (50)
Guidance (score distillation)
MVDream – best quality, multi-view (recomme
Generate 3D on GPU
GALLERY – CLICK ANY MODEL TO RENDER IT
a pink frosted fried ring pastry with
community MVDREAM · seed 0 · 500
2 Real-time novel-view renderer · “a round fluffy night bird with big yellow eyes and ear tufts”
Photo shoot · make the judge see an owl
Take photo
Send 3 photos to the judge
Keep auto-camera verdict
studio lighting
light direction
light height
backdrop
studio light
auto-rotate
splat scale
1.00
render
colour
reset view
Save image
.ply
CUDA rasterizer snapshots during SDS optimisation (front · side · back · side)
final
3 Optimisation monitor
SDS loss ½‖w(t)(ε̂−ε)‖² per step
1e5
1e4
1e3
1e2
step 1
500
Gradient norm reaching each Gaussian parameter (log)
1e5
1e1
1e-3
1e-7
step 1
499
position
colour
opacity
scale
rotation
Number of Gaussians (densify / prune)
2.39e+5
1.20e+5
0
step 1
500

```
60 FPS (rAF)  ·  GPU 7.40 ms
221,955 Gaussians  ·  710×532
sort 6 ms  ·  NVIDIA Corporation, NVIDIA A100-PCIE-40GB/PCIe
```

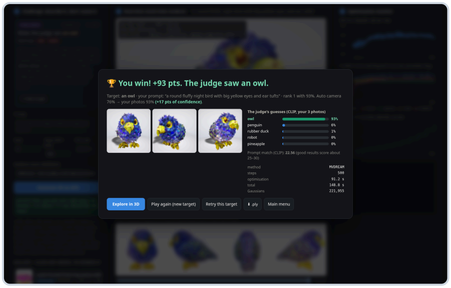
Write a prompt that describes it without those words. When your 3D model is done, you get a preview verdict from 8 automatic views, then a **photo shoot**: light and frame the model yourself, take up to 3 photos, and the CLIP judge guesses which of 20 objects it is from your photos. If it guesses yours, you win points.

Stage your model for the judge: orbit to the best angle, move the light, pick a backdrop. Take up to 3 photos, then send them. **The judge sees only your photos.**

● 8 GPUs free - your job starts right away. ETA: finishes ≈ 3:41 AM (in ~2.1 min; MVDREAM, 500 steps).

Public demo: the server runs one job on every GPU that nobody else is using (the count above updates live) (about 2.1 min per model on an A100); other prompts wait in a queue and start automatically. Each session can have one job in progress at a time, max 500 SDS steps, no mesh export. Every finished model is added to the public gallery below.

Photo shoot: studio lighting, light direction and height, backdrop; 3 photo slots.


Explore in 3D
Play again (new target)
Retry this target
.ply
Main menu

🏆 You win! +93 pts. The judge saw an owl.

Target: **an owl** · your prompt: "a round fluffy night bird with big yellow eyes and ear tufts" · rank 1 with 93%. Auto camera 76% → your photos 93% **(+17 pts of confidence)**.

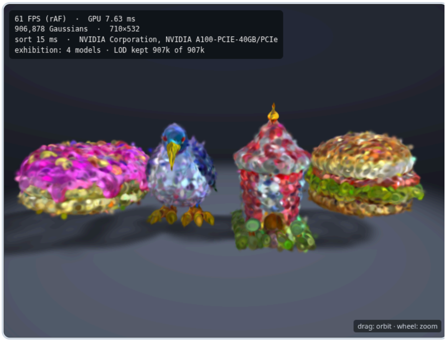
The judge's guesses (CLIP, your 3 photos)

owl	93%
penguin	6%
rubber duck	1%
robot	0%
pineapple	0%

Prompt match (CLIP): **22.56** (good results score about 25–30)

method	MVDREAM
steps	500
optimisation	91.2 s
total	148.8 s
Gaussians	221,955

End state: verdict from the player's photos, points, comparison with the preview.


drag: orbit · wheel: zoom

```
61 FPS (rAF)  ·  GPU 7.63 ms
906,878 Gaussians  ·  710×532
sort 15 ms  ·  NVIDIA Corporation, NVIDIA A100-PCIE-40GB/PCIe
exhibition: 4 models · LOD kept 907k of 907k
```

Exhibition: trophies and examples on one lit stage, one sort, LOD budget.

3. Performance: the A2 target (≥ 30 FPS) holds in every playable setting

setting (headless Chrome, WebGL2, A100)	Gaussians	canvas	FPS (rAF, capped 60)	GPU ms (splat pass)	sort ms (worker)
single model, raw splats	220,296	810x607	61	6.9	4
single model, studio lighting + stage	220,296	810x607	61	6.87	4
single model x8 replicas, studio	1,762,368	810x607	18	51.85	30
exhibition 4 models, studio	1,100,201	810x607	56	14.08	18
exhibition 8 models, LOD 1.2M	1,200,000	810x607	61	11.55	18
exhibition 8 models, LOD 2.4M	2,147,257	810x607	40	20.19	34

Lighting is computed per Gaussian in the vertex shader, so it costs nothing measurable (6.9 → 6.87 ms); the frame is bound by the number of Gaussians. Every setting a player can reach stays at ≥ 40 FPS (worst 20.2 ms GPU). The A3 stress case of 8 replicas stacked on one spot (1,762,368 Gaussians, heavy overdraw) drops to 18 FPS; this is why the exhibition keeps each model's most opaque Gaussians within an LOD budget (default 1.2 M → 61.0 FPS). `scripts/bench_beta.py` → `evidence/beta_render.csv`.

4. Beta release notes

New features: studio lighting from per-Gaussian normals (toggle, light direction and height); four backdrops (studio dark, studio light, plain white, plain black) with ground plane and soft shadow; **photo shoot** phase in every Challenge round with a CPU CLIP judge endpoint (`/api/judge_photos`, about 1 s for 3 photos, never takes a GPU); points per win and a points total in the HUD; **Exhibition** menu mode (trophies first, then your models and examples; 2–8 models; LOD budget 0.6/1.2/2.4 M; two rows above 4 models); camera target support in the viewer; the result card shows the player's photos and the change against the preview. **Stability:** leaving a round mid-shoot (menu, Esc, new job) keeps the preview verdict instead of losing the round; the photo judge is loaded at server start; photos are size-checked and one judgement per visitor runs at a time; the Beta server moved to *hermes* after *homer* rebooted.

Known issues: normals from the shortest axis are noisy on fuzzy, blob-like Gaussians, so lighting can look speckled; the shadow uses Gaussian centres only (no self-shadowing); the judge can be gamed by backdrop choice (studio dark vs. white changes the score), which is part of the game but makes scores depend on the look; the quick-tunnel URL changes when the tunnel restarts; points and trophies live in `sessionStorage` (lost with the tab) until the SQLite leaderboard in week 12; the exhibition downloads every model's full PLY before LOD (about 10–15 MB each).

5. Updated production schedule (weeks from Sep 1; final deadline in week 15)

Week	Engineering / content	Milestone
9 · Oct 26–Nov 1	Alpha: game loop, CLIP judge, render modes, offline jobs, watchdog	Alpha v0.2 (A4)
10–11 · Nov 2–15	**Beta (this build):** lit splats (normals, wrap diffuse, specular, rim), stage + soft shadow, photo shoot inside the Challenge round, CPU photo judge, exhibition with LOD; real MVDream rounds; render benchmark	Beta v0.3 live + tagged (this report)
12 · Nov 16–22	Persistence (SQLite: rounds, points, leaderboard), moderation of the gallery, stable hosting (named tunnel); photo-judge robustness (prompt ensembles, more targets)	**Integration:** leaderboard + photo points; classmate play-test
13 · Nov 23–29 (Thanksgiving)	Fix play-test issues; per-pixel shading option; mobile layout of the photo shoot; GPU sort for very large exhibitions	Play-test triage
14 · Nov 30–Dec 6	**Feature freeze:** fix top issues only; load test with parallel players; record final video	Release candidate v1.0-rc
15 · Dec 7–13	**BUFFER WEEK:** bug fixing, documentation and polish only; no new features	v1.0 + final submission

green done · purple integration · orange buffer week (no new features). Run: `bash scripts/setup_env.sh`, then `bash scripts/serve_public.sh` (or `python server.py`); Linux + NVIDIA GPU ≥ 12 GB.